Midterm 2, Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

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Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. (a) (2 points) Give an example or briefly explain why the following is impossible: a random variable X such that $\mathbb{E}X = 3$ and $\mathbb{E}(X^2) = 3$.

   Solution: this is impossible since $\mathrm{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = 3 - 9 < 0$, but $\mathrm{Var}(X) = \mathbb{E}(X - \mathbb{E}X)^2 \geq 0$ always.

   (b) (2 points) Give an example or briefly explain why the following is impossible: a random variable X such that $\mathbb{E}X = 3$ and $\mathrm{Var}\, X = 3$.

   Solution: for example, Poisson(3)

   (c) (2 points) If random variables Y and Z have binomial distributions (with some parameters), can $Y + Z$ have geometric distribution? Justify your answer.

   Solution: this is impossible since $Y+Z$ attain finitely many different values (unlike geometric random variable)

   (d) (2 points) Let X be a Binomial random variable such that $\mathbb{E}(X) = 4$ and $\mathrm{Var}(X) = 2$. Find the parameters of X.

   Hint: to recall the formulas for the mean and variance of Binomial, think of Binomial random variable as a sum of independent Bernoulli random variables.

   Solution: $np = 4$ and $np(1-p) = 2$, hence, $1 - p = 1/2$ and $p = 1/2$. So, $n = 8$.

   (e) (2 points) Let $X \sim \mathrm{Unif}[0, 2]$ (continuous random variable). Compute $\mathrm{Var}(e^X)$

   Hint: PDF $f_X(x) = c$ when $0 \leq x \leq 2$ and zero otherwise.

   Solution: $\mathrm{Var}(Y) = \mathbb{E}Y^2 - (\mathbb{E}Y)^2$, here, $Y = e^X$ and $Y^2 = e^{2X}$.

   $\mathbb{E}(Y) = \mathbb{E}(e^X) = \int_0^2 e^t \frac{1}{2} dt = (e^2 - 1)/2$.

   $\mathbb{E}(Y^2) = \mathbb{E}(e^{2X}) = \int_0^2 e^2 t \frac{1}{2} dt = (e^4 - 1)/4$.

   So, $\mathrm{Var}(e^X) = (e^4 - 1)/4 - (e^2 - 1)^2/4$.

2. Let (X, Y) be a random point which is distributed uniformly in the set $\{(x, y) : x, y \text{ integers}, -2 \leq x \leq 2, -3 \leq y \leq 3\}$.

   (a) (2 points) Find the marginal PMFs of X and of Y.

   (b) (2 points) Find $\text{Var}(X - 2Y)$.

   (c) (3 points) Are $X + Y$ and $X - Y$ independent? Prove your claim.

   (d) (3 points) Compute explicitly $\mathbb{E}[X \sin(\pi Y^{2018})]$

Solution: (a) X and Y are discrete uniform in $\{-2, -1, \cdots, 2\}$ and $\{-3, \cdots, 3\}$ respectively.

(b)

$$\text{Var}(X - 2Y) = \text{Var}(X) + \text{Var}(2Y) = \text{Var}(X) + 4\,\text{Var}(Y) = \frac{4 \cdot 6}{12} + \frac{6 \cdot 8}{12} = 6$$

(c) No. Intuitively if we know that $X + Y = 5$, then we know that $X = 2$ and $Y = 3$, and hence $X - Y = -1$. To prove that they are not independent we have to find out two numbers m, n such that

$$\mathbb{P}(X + Y = m, X - Y = n) \neq \mathbb{P}(X + Y = m)\mathbb{P}(X - Y = n).$$

From the intuition, choose $m = 5$, $n = -1$. We have

$$\mathbb{P}(X + Y = 5, X - Y = -1) = \mathbb{P}(X = 2, Y = 3) = \frac{1}{35}.$$

And

$$\mathbb{P}(X + Y = 5) = \mathbb{P}(X = 2, Y = 3) = \frac{1}{35},$$

$$\mathbb{P}(X - Y = -1) \leq 1 - \mathbb{P}(X = 0, Y = 0) < 1.$$

Hence

$$\mathbb{P}(X + Y = 5, X - Y = -1) > \mathbb{P}(X + Y = 5)\mathbb{P}(X - Y = -1).$$

(d) Since X and Y are independent

$$\mathbb{E}[X \sin(\pi Y^{2016})] = \mathbb{E}(X)\mathbb{E}[\sin(\pi Y^{2016})] = 0$$

3. (10 points) The number of cars that arrive in a certain mechanic shop is a Poisson random variable with the parameter $\lambda = 1$. For $k \geq 1$, given that exactly k cars arrive, the number of hours T that it takes to repair all of the cars is a (discrete) uniform random variable taking values in $\{1, 2, \dots, k+1\}$.

   (a) Find the probability that 1 hour will be enough for all the repairs.

   (b) Find the expectation $\mathbb{E}(T)$.

   *(c) [Optional] Simplify both of the answers so that no sums left.
   Hint: In the first part you might find the following formula useful:

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e.$$

   In the second part you can observe the sum as an expectation of $f(X)$, where f is a very simple function of a known random variable.

Solution:

   (a) Use the total probability theorem. Let X be the number of cars that arrived. Then

$$\mathbb{P}(T \leq 1) = \sum_{k=0}^{\infty} \mathbb{P}(T \leq 1 | X = k)\mathbb{P}(X = k).$$

   If $X = 0$ then $T = 0$ so $\mathbb{P}(T \leq 1 | X = 0) = 1$. For $k \geq 1$ we have $\mathbb{P}(T \leq 1 | X = k) = \frac{1}{k+1}$. Then

$$\mathbb{P}(T \leq 1) = e^{-1} + \sum_{k=1}^{\infty} \frac{1}{k+1} \frac{e^{-1}}{k!} = e^{-1}\Big(1 + \sum_{k=1}^{\infty} \frac{1}{(k+1)!}\Big) = e^{-1}\Big(e - \frac{1}{1!}\Big) = 1 - e^{-1}.$$

   (b) Given $X = k$ the conditional expectation $\mathbb{E}(T|X = k) = (k+2)/2$, for $k \geq 1$ and $\mathbb{E}(T|X = 0) = 0$. We can use the formula

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{\infty} \mathbb{E}(T|X = k)\mathbb{P}(X = k) = \sum_{k=1}^{\infty} \frac{k+2}{2} \frac{e^{-1}}{k!} \\
&= \sum_{k=0}^{\infty} \frac{k+2}{2} \frac{e^{-1}}{k!} - \frac{e^{-1}}{2} = \mathbb{E}((X+2)/2) - \frac{e^{-1}}{2} = \frac{1+2}{2} - \frac{e^{-1}}{2} = \frac{3 - e^{-1}}{2}.
\end{aligned}$$

4. (10 points) You are attending a class where students show up T minutes late, where T is exponentially distributed with parameter 1. The instructor decides to take attendance and deduct points based on how late a student arrives. If a student arrives less than 1 minute late, the instructor takes away 0 points. If a student arrives between 1 and 3 minutes late, 5 points are deducted. If a student arrives between 3 and 6 minutes late, 10 points are deducted. Any student that arrives more than 6 minutes late is deducted 20 points. Compute the expected number of points deducted for a given student.

Solution: Let D be a random variable equal to the number of points deducted. Possible values for D are $0, 5, 10, 20$, so it is a discrete random variable.

$\mathbb{E}(D) = 0 \cdot \mathbb{P}(D = 0) + 5 \cdot \mathbb{P}(D = 5) + 10 \cdot \mathbb{P}(D = 10) + 20 \cdot \mathbb{P}(D = 20) = 0 + 5 \cdot \mathbb{P}(1 < T \leq 3) + 10 \cdot \mathbb{P}(3 < X \leq 6) + 20 \cdot \mathbb{P}(X > 6)$

As we computed in class, $\mathbb{P}(T > x) = e^{-x}$ (not hard to check by the definition of PDF of exponential random variable).

Then, $\mathbb{P}(1 < T \leq 3) = \mathbb{P}(1 < T) - \mathbb{P}(T > 3) = e^{-1} - e^{-3}$ and $\mathbb{P}(3 < T \leq 6) = e^{-3} - e^{-6}$.

Answer: $\mathbb{E}(D) = 5(e^{-1} - e^{-3}) + 10(e^{-3} - e^{-6}) + 20e^{-6} = 5e^{-1} + 5e^{-3} + 10e^{-6}$.

5. (10 points) A box contains n balls, exactly one of which is red and one of which is yellow. We draw a ball uniformly at random observe the color, return it back to the box and repeat this indefinitely. Let X be the number of draws we had to do to draw the yellow ball, and let Y be the number of draws we had to do to draw the red ball.

   (a) Compute $\mathbb{E}[X+Y]$. Are X and Y independent?

   (b) Compute all the possible values that pair (X, Y) can have, and the joint PMF of (X, Y).

   Solution:

   (a) Observe that the outcome of each draw is independent, and each time we have probability $1/n$ to draw yellow. Therefore, X has geometric distribution with parameter $1/n$. This is also true for Y so

   $$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 2n.$$

   Variables are not independent, since they can't have equal values. In other words, the range of Y is $\{1, 2, \ldots\}$ and yet given $X = k$, the conditional probability that $Y = k$ is zero (for any k).

   (b) The pair (X, Y) can attain any value (x, y), where x and y are different positive integers. For $x < y$ to have $X = x$ and $Y = y$ we need to avoid drawing either yellow or red in the first $x-1$, draw yellow at x-th draw, avoid drawing the red ball in the next $y-x-1$ draws and the red in the y-th draw. This has the probability

   $$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y) = \left(\frac{n-2}{n}\right)^{x-1} \cdot \frac{1}{n} \cdot \left(\frac{n-1}{n}\right)^{y-x-1} \cdot \frac{1}{n} = \frac{(n-2)^{x-1}(n-1)^{y-x-1}}{n^y}.$$

   In the same way for $x > y$ we have

   $$p_{X,Y}(x, y) = \frac{(n-2)^{y-1}(n-1)^{x-y-1}}{n^x}.$$